

SHAIK MOHAMMAD ASHIK

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Technical Summary

Systems-focused Electrical and Electronics Engineering undergraduate specializing in the intersection of firmware concepts, Python scripting, and real-time telemetry instrumentation. Experienced in developing responsive software interfaces for simulated physical systems, state tracking, and electrical parameter rendering. Proven competence in structured engineering problem-solving, mixed-signal analysis, and hardware-software co-design paradigms.

Core Competencies

Languages & Scripts: C, Python, JavaScript (ES6+), HTML5, CSS3

Embedded & Hardware Systems: Microcontroller Architecture, Mixed-Signal Sensor Interfacing, Telemetry Ingestion, Digital Logic Design

Software Architectures: Client-Side State Management, Asynchronous Event Handling, DOM Architecture, Data Visualization Runtimes

Developer Ecosystem: Git, GitHub, Linux CLI, Automation Scripts

Education

Vasireddy Venkatadri Institute of Technology (VVIT)

Andhra Pradesh, India *Rel-*

Bachelor of Technology (B.Tech) – Electrical and Electronics Engineering (CGPA: 7.52 / 10.00) 2023 – 2027 (Expected)
evant Coursework: Power Electronics, Linear Control Systems, Microprocessors & Microcontrollers, Analog/Digital Circuit Design, Embedded Systems Design Foundations.

NxtWave CCBP 4.0 Academy

Remote

Advanced Specialization in Systems and Front-End Development

Apr 2023 – Apr 2025

Systems Engineering Projects

Smart Energy Monitoring & Telemetry Dashboard | *JS (ES6), HTML5 Canvas, CSS Grid*

- **Problem:** Standard electrical monitoring setups lack responsive, low-overhead visual client engines capable of processing simulated multi-channel sensor arrays without blockages.
- **Architecture:** Designed an isolated client-side engine utilizing an asynchronous data generator to simulate raw ADC telemetry outputs (Voltage, Current, Active Power, Power Factor) mapped to standard grid constraints.
- **Implementation:** Engineered a zero-dependency data translation layer converting raw numeric streams into real-time visual vectors; developed localized mathematical tracking algorithms to compute instantaneous values and historical peak consumption parameters.
- **Engineering Trade-offs:** Prioritized native Web APIs over heavy third-party plotting frameworks to minimize client-side memory usage and maximize execution speed on low-power devices.

Embedded Telemetry Interface & Signal Logger | *Python, C Foundations, OOP*

- **Problem:** Lack of efficient programmatic frameworks to parse, log, and isolate out-of-bounds anomalies in serialized hardware sensor transmissions.
- **Architecture:** Developed a modular Python application structured to handle continuous data processing, error detection, and programmatic logging of simulated sensor anomalies.
- **Implementation:** Coded robust validation logic using Python data structures (dictionaries, sets) to rapidly index, match, and categorize received telemetry codes against expected nominal operation parameters.

Certifications & Job Simulations

- **Job Simulation:** J.P. Morgan Investment Banking Institutional Analysis Virtual Experience (Forage) – Evaluated financial modeling mechanics and corporate data analysis workflows.
- **Technical Accreditations:** CCBP 4.0 Academy-Genius Certification (Deep Dive in Functional Programming and Client Architecture Verification).